* Learning Rate

* Step:-

→ Start with a high learning rate and divide it by 10 when reaching a plateau.

* Cosine

→ Continous Decay

→ Given the no. of training epoch E, the starting learning rate lr0 and ending learning rate lrE, the learning rate epoch e is given by:

$$lr_e = lr_E + \frac{1}{2}(lr_0 - lr_E)\left(1 + \cos\left(\frac{e\pi}{E}\right)\right)$$

→ compare to step scedular it only require two hyperparameter (starting and ending).

* Linear

→ continous decay of learning rate that follows a linear function.

→ the starting & ending is

$$lr_e = lr_E + (lr_0 - lr_E)\left(1 + \frac{e}{E}\right)$$

* Warmer-up

→ Start with a small learning rate for a few steps before growing & progressively decaying

→ Useful for large networks where poor initialization and high learning rates at the begining might slow down convergence.

* One Cycle:-

→ defined on batch steps.

→ start with a small learning rate that grows until a maximum is reached after which decay starts.

$$lr_i = \begin{cases} lr_{max} + (lr_0 - lr_{max})\left(1 - i/pI\right) & \text{if } i < pI \\ lr_{min} + (lr_0 - lr_{min})\left(1 - \dfrac{i - pI}{I - pI}\right) & \text{if } i \geq pI \end{cases}$$

* Label Smoothing

→ Label Smoothing is a regulization technique used to prevent overfitting in classification tasks.

→ Normally, when using softmax with cross-entropy loss, the model tries to make the predicted probability for the correct class to close to 1 as possible and all other to zero.

→ This can lead to overconfidence & poor generalization.

→ Instead of using hard label like:-

* correct class = 1
* All other = 0

→ We soften the label a little using a small value ε. For a classification problem with C class

* correct class =

$$y(i) = 1 - \varepsilon$$

* All other class

$$y(i) = \frac{\varepsilon}{C-1}$$

This means the model doesn't have to be 100% confident making it more robust & less likely to overfit.